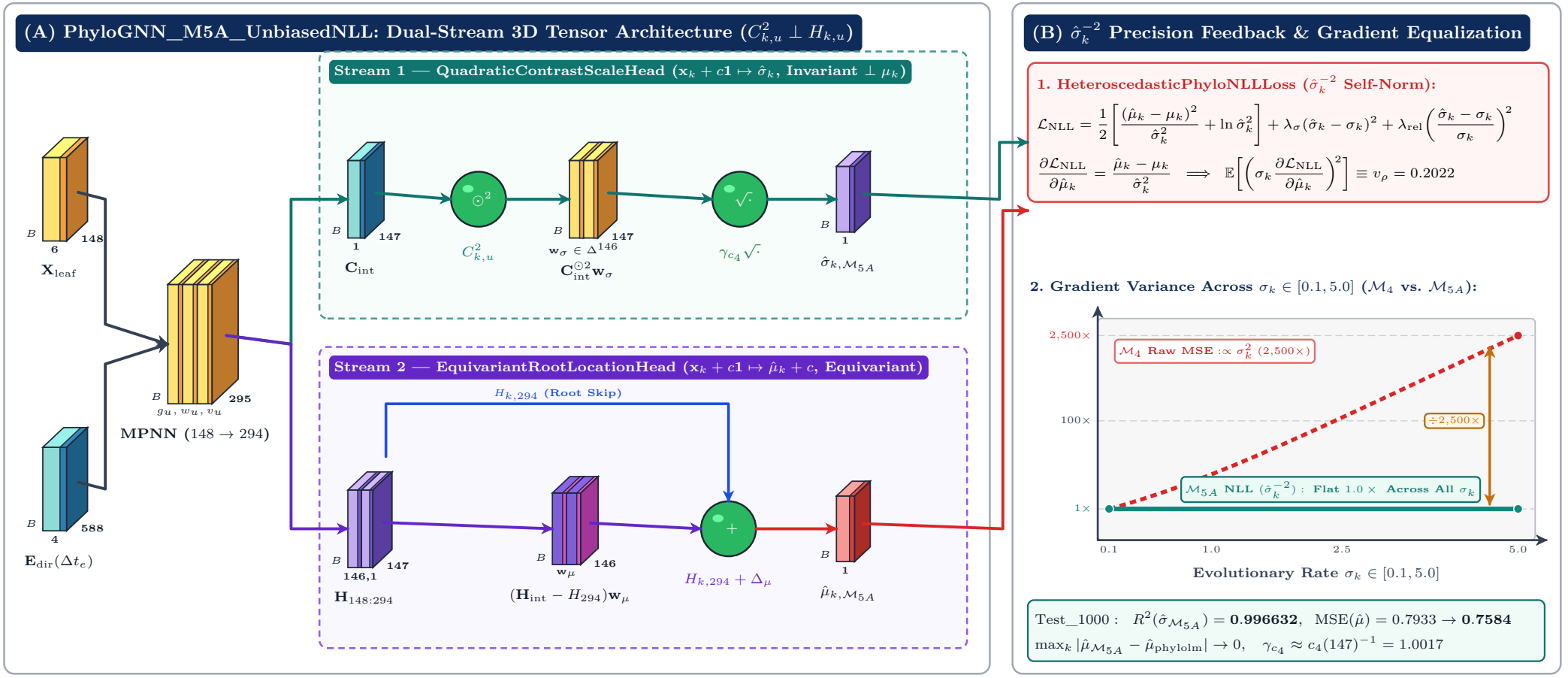

(A) PhyloGNN_M5A_UnbiasedNLL: Dual-Stream 3D Tensor Architecture ($C^2_{k,u} \perp H_{k,u}$)
Stream 1 — QuadraticContrastScaleHead ($x_k + c1 \mapsto \hat{\sigma}_k$, Invariant $\perp \mu_k$)
X_{leaf}
MPNN (148 → 294)
g_u, w_u, v_u
$E_{dir}(\Delta t_e)$
C_{int}
$C^2_{k,u}$
$C^{\odot 2}_{int} w_\sigma$
$\gamma_{c_4}\sqrt{\cdot}$
$\hat{\sigma}_{k,\mathcal{M}_{5A}}$
Stream 2 — EquivariantRootLocationHead ($x_k + c1 \mapsto \hat{\mu}_k + c$, Equivariant)
$H_{k,294}$ (Root Skip)
$H_{148:294}$
$(H_{int} - H_{294})w_\mu$
$H_{k,294} + \Delta_\mu$
$\hat{\mu}_{k,\mathcal{M}_{5A}}$
(B) $\hat{\sigma}_k^{-2}$ Precision Feedback & Gradient Equalization
1. HeteroscedasticPhyloNLLLoss ($\hat{\sigma}_k^{-2}$ Self-Norm):
$\mathcal{L}_{NLL} = \frac{1}{2}\left[\frac{(\hat{\mu}_k - \mu_k)^2}{\hat{\sigma}_k^2} + \ln \hat{\sigma}_k^2\right] + \lambda_\sigma(\hat{\sigma}_k - \sigma_k)^2 + \lambda_{rel}\left(\frac{\hat{\sigma}_k - \sigma_k}{\sigma_k}\right)^2$
$\frac{\partial \mathcal{L}_{NLL}}{\partial \hat{\mu}_k} = \frac{\hat{\mu}_k - \mu_k}{\hat{\sigma}_k^2} \implies \mathbb{E}\left[\left(\sigma_k \frac{\partial \mathcal{L}_{NLL}}{\partial \hat{\mu}_k}\right)^2\right] \equiv v_\rho = 0.2022$
2. Gradient Variance Across $\sigma_k \in [0.1, 5.0]$ ($\mathcal{M}_4$ vs. $\mathcal{M}_{5A}$):
2,500×
100×
1×
$\mathcal{M}_4$ Raw MSE $\propto \sigma_k^2$ (2,500×)
÷2,500×
$\mathcal{M}_{5A}$ NLL ($\hat{\sigma}_k^{-2}$): Flat 1.0 × Across All σ_k
0.1
1.0
2.5
5.0
Evolutionary Rate $\sigma_k \in [0.1, 5.0]$
Test_1000 : $R^2(\hat{\sigma}_{\mathcal{M}_{5A}}) = \mathbf{0.996632}$, MSE($\hat{\mu}$) = 0.7933 → **0.7584**
$\max_k |\hat{\mu}_{\mathcal{M}_{5A}} - \hat{\mu}_{phylolm}| \to 0$, $\gamma_{c_4} \approx c_4(147)^{-1} = 1.0017$